Experiment 6: Total Cost with Discount and Tax

Aim

To calculate the total cost of a customer's purchase by considering item prices, quantities, discount percentage, and tax percentage using arithmetic operations.

Algorithm

1. Start the program.
2. Create lists containing item prices and quantities.
3. Initialize the discount rate and tax rate.
4. Calculate the cost of each item by multiplying its price by its quantity.
5. Calculate the subtotal by adding the costs of all items.
6. Calculate the discount amount and subtract it from the subtotal.
7. Calculate the tax on the discounted amount.
8. Add the tax to the discounted amount to obtain the final cost.
9. Display the subtotal, discount, tax, and total cost.
10. Stop the program.

Code

```
prices = [100, 200, 50]

quantities = [2, 1, 3]

discount_rate = 10

tax_rate = 5

subtotal = 0

for i in range(len(prices)):

  subtotal += prices[i] * quantities[i]

discount = subtotal * discount_rate / 100

after_discount = subtotal - discount

tax = after_discount * tax_rate / 100

total_cost = after_discount + tax

print("Subtotal:", subtotal)
```

```
print("Discount:", discount)

print("Tax:", tax)

print("Total :", total_cost)
```

Input

Item prices = [100, 200, 50]

Quantities = [2, 1, 3]

Discount rate = 10%

Tax rate = 5%

Output

```
========== RESTART: C:/Users/arunp/OneDrive/Documents/FDS/EXP 5.py.py ==========
Enter the number of items: 3
Enter price of item 1: 100
Enter quantity of item 1: 2
Enter price of item 2: 200
Enter quantity of item 3: 3
Enter price of item 3: 50
Enter quantity of item 3: 3
Enter discount rate (%): 10
Enter tax rate (%): 5
Subtotal: 950.0
Discount: 95.0
Tax: 42.75
Total cost: 897.75
```

Result

Thus, the total cost of the customer's purchase was calculated as 519.75, including a 10% discount and 5% tax on the discounted amount.